

Evaluating the Performance of Object Detection Models for Accurate Sri Lankan Traffic Sign Detection



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Course & Batch: HND in Computing - 22

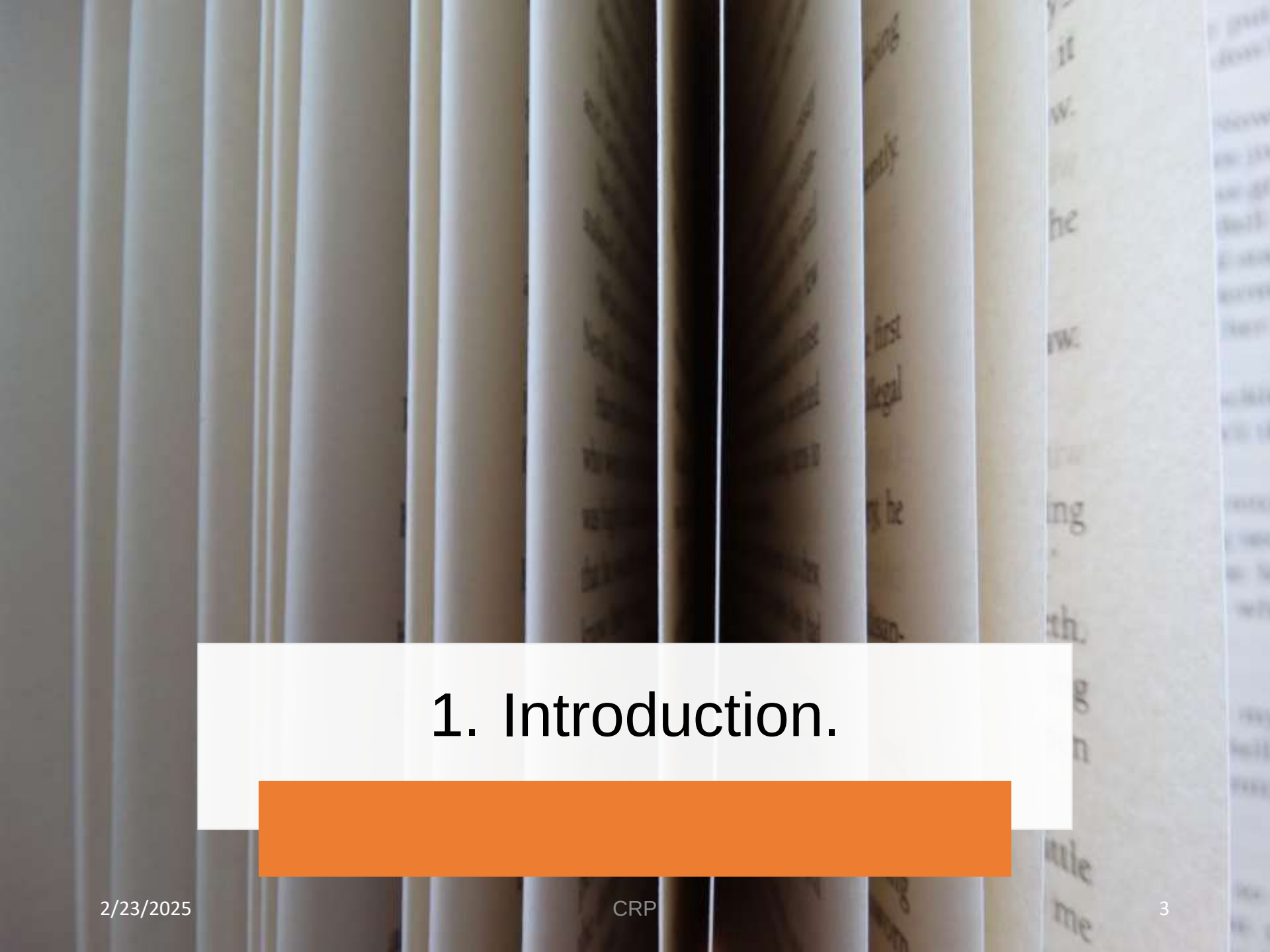
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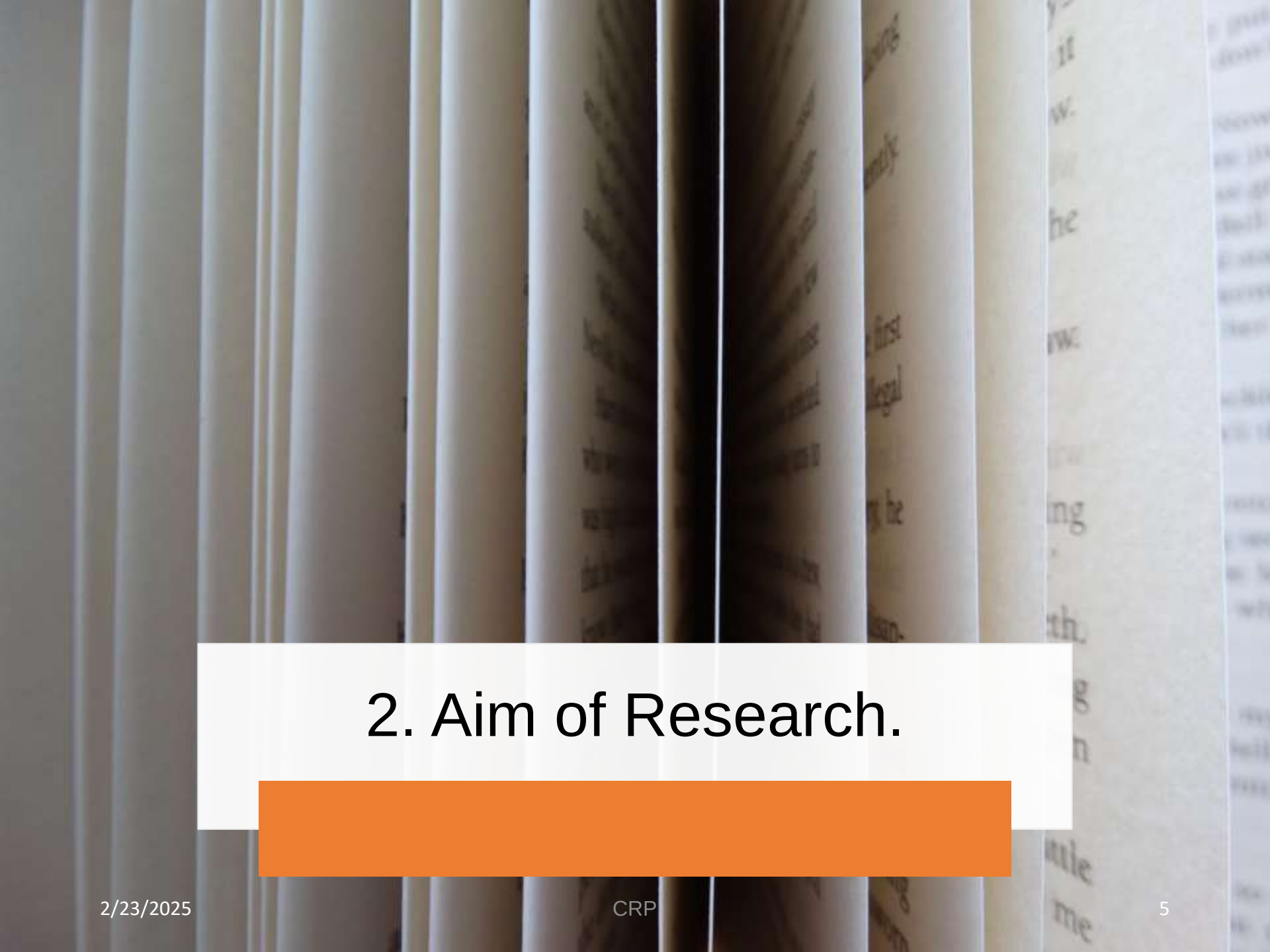
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The background of the slide is a close-up, slightly blurred image of the pages of an old book. The pages are yellowed with age, and some text is visible, though not legible. Overlaid on the bottom half of the image is a white rectangular box. Inside this box, the text '1. Introduction.' is written in a large, black, sans-serif font. Below the white box is a solid orange horizontal bar.

1. Introduction.

Introduction

- Importance of traffic sign detection in AI and computer vision.
- Relevance to Sri Lankan road systems.
- Comparing YOLOv8, Faster R-CNN, and SSD using mAP.



2. Aim of Research.

Aim

- To compare the performance of YOLOv8, Faster R-CNN, and SSD for Sri Lankan traffic sign detection and identify the model with the highest mAP, contributing to improved automated traffic sign recognition and enhanced road safety in Sri Lanka using AI.



3. Research Question

Research Question

- **Research Question:** Which object detection model (YOLOv8, Faster R-CNN, or SSD) achieves the highest accuracy in detecting Sri Lankan traffic signs, as measured by mean Average Precision (mAP)?
- **Dependent Variables:** Mean Average Precision (mAP) (the accuracy of the models).
- **Independent Variable:** The object detection model used (YOLOv8, Faster R-CNN, SSD).

The background of the slide is a photograph of a bookshelf. The books are arranged vertically, and their spines are visible. The text on the spines is mostly illegible due to the shallow depth of field, but some words like "first", "legal", and "he" can be seen. A white rectangular box is overlaid on the lower half of the image, containing the section header. Below the white box is a solid orange horizontal bar.

4. Objective of Research

Objective & Sub objectives

Sub Objectives:

- Review existing research on object detection models for traffic sign detection.
- Understand the strengths and weaknesses of YOLOv8, Faster R-CNN, and SSD.

Objective

- To conduct a comprehensive comparison of existing object detection models (YOLOv8, Faster R-CNN, and SSD) in terms of their suitability for Sri Lankan traffic sign detection.
- To evaluate the performance of these models using the mean Average Precision (mAP) metric on a Sri Lankan traffic sign dataset.
- To compare and analyze the results, identifying the most effective model for accurate traffic sign detection in Sri Lanka.



5. Scope of Research

Scope

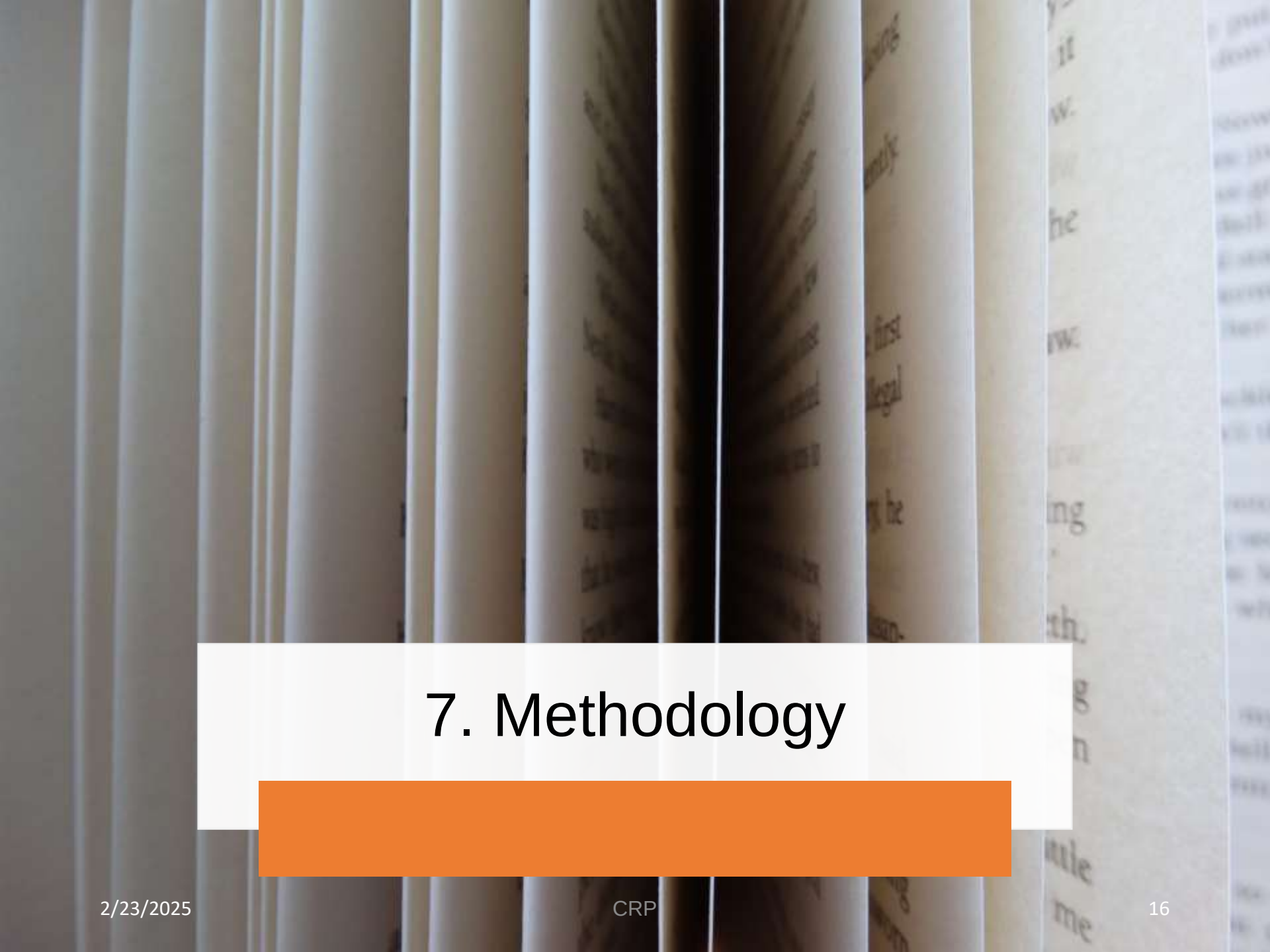
- **Focus Area:** Evaluating object detection models for Sri Lankan traffic sign detection.
- **Models Considered:** YOLOv8, Faster R-CNN, and SSD.
- **Dataset:** A dataset containing Sri Lankan traffic signs.
- **Evaluation Metric:** Mean Average Precision (mAP) as the primary performance measure.

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6.Limitations of Research

Limitations

- The study focuses only on model evaluation, not implementation or fine-tuning.
- Research is limited to Sri Lankan traffic signs and may not generalize to other regions.



7. Methodology

Methodology

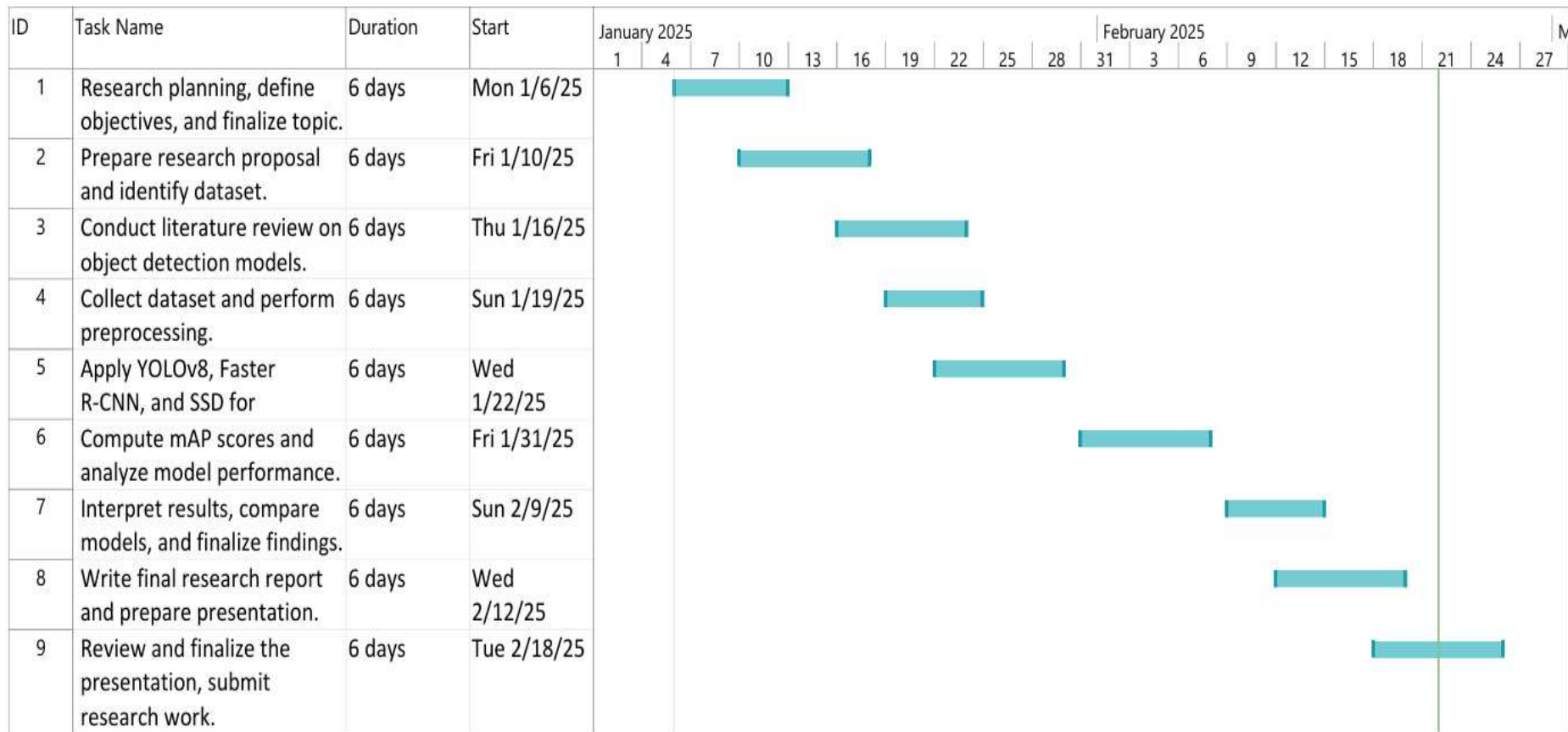
- **Research Design:** Quantitative research.
- **Data Collection Method:** Secondary dataset (Sri Lankan traffic sign dataset).
- **Evaluation Approach:** Analyze the performance of YOLOv8, Faster R-CNN, and SSD. Use mean Average Precision (mAP) as the key evaluation metric.
- **Comparison & Analysis:** Compare the models based on mAP scores. Identify the most effective model for Sri Lankan traffic sign detection.

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Literature Review

Title	Aim	Results	Methodology	Research Gap
Performance Comparison of Object Detection Models for Road Sign Detection Under Different Conditions – 2024 (Fatima, Z. T. M. M. H. V. R. R. T. a. R. R., 2024. Performance Comparison of Object Detection Models for Road Sign Detection Under Different Conditions. International Journal of Advanced Computer Science and Applications, 15(12).)	state-of-the-art object detection models (YOLOv8, YOLOv9, RTMDet, Faster R-CNN, RetinaNet) for road sign detection.	YOLOv9 and YOLOv8 performed best in real-time detection, while RTMDet provided a balance of speed and accuracy. Faster R-CNN and RetinaNet had better precision but were slower.	Models were fine-tuned and evaluated on a large dataset using mAP, FPS, and total loss as key metrics. Training involved data augmentation and hyperparameter tuning .	Focused on general road sign detection but did not evaluate models on region-specific datasets.
A Traffic-Sign Detection Algorithm Based on Improved Sparse R-CNN – 2017 (Jinghao Cao, J. Z. X. J., 2021. A Traffic-Sign Detection Algorithm Based on Improved Sparse R-cnn. IEEE Access.)	To enhance Sparse R-CNN for traffic sign detection using an improved backbone and attention mechanism.	Achieved improved mAP and robustness compared to traditional Sparse R-CNN. Res2Nest and improved DIIH enhanced detection, especially for small signs.	Proposed an improved ResNest-based backbone (Res2Nest) and Dynamic Instance Interactive Head (DIIH) for better multi-scale feature representation and accuracy. Evaluated using the TT100K dataset .	Needs optimization for lower computational cost while maintaining accuracy.
Evaluating and Benchmarking Object Detection Models for Traffic Sign Detection – 2022 (Mishra, A. K. A. M. S. A. K. Z. J. S. S. a. T. A., 2022. Evaluating and Bench-marking Object Detection Models for Traffic Sign and Traffic Light Datasets, s.l.: Computer Vision Foundation.)	To benchmark object detection models (Faster R-CNN, YOLOv3, SSD, RetinaNet, CenterNet, Cascade R-CNN) for traffic sign detection across multiple datasets.	CenterNet and Cascade R-CNN performed best overall, with CenterNet excelling in small object detection. Faster R-CNN and YOLOv3 provided competitive results in specific datasets.	Models were evaluated using precision, recall, and mean average precision (mAP) on different datasets, including TT100K and GTSD .	Existing models struggle under adverse weather and occlusions.

Timeline





Cost & Benefits

Description	Cost
Power Consumption	1,800 LKR
Computational resources	4,000 LKR
Cloud services	5,500 LKR
Infra Structure	1,500 LKR
Manpower	3,700 LKR
Total Cost	16,500 LKR

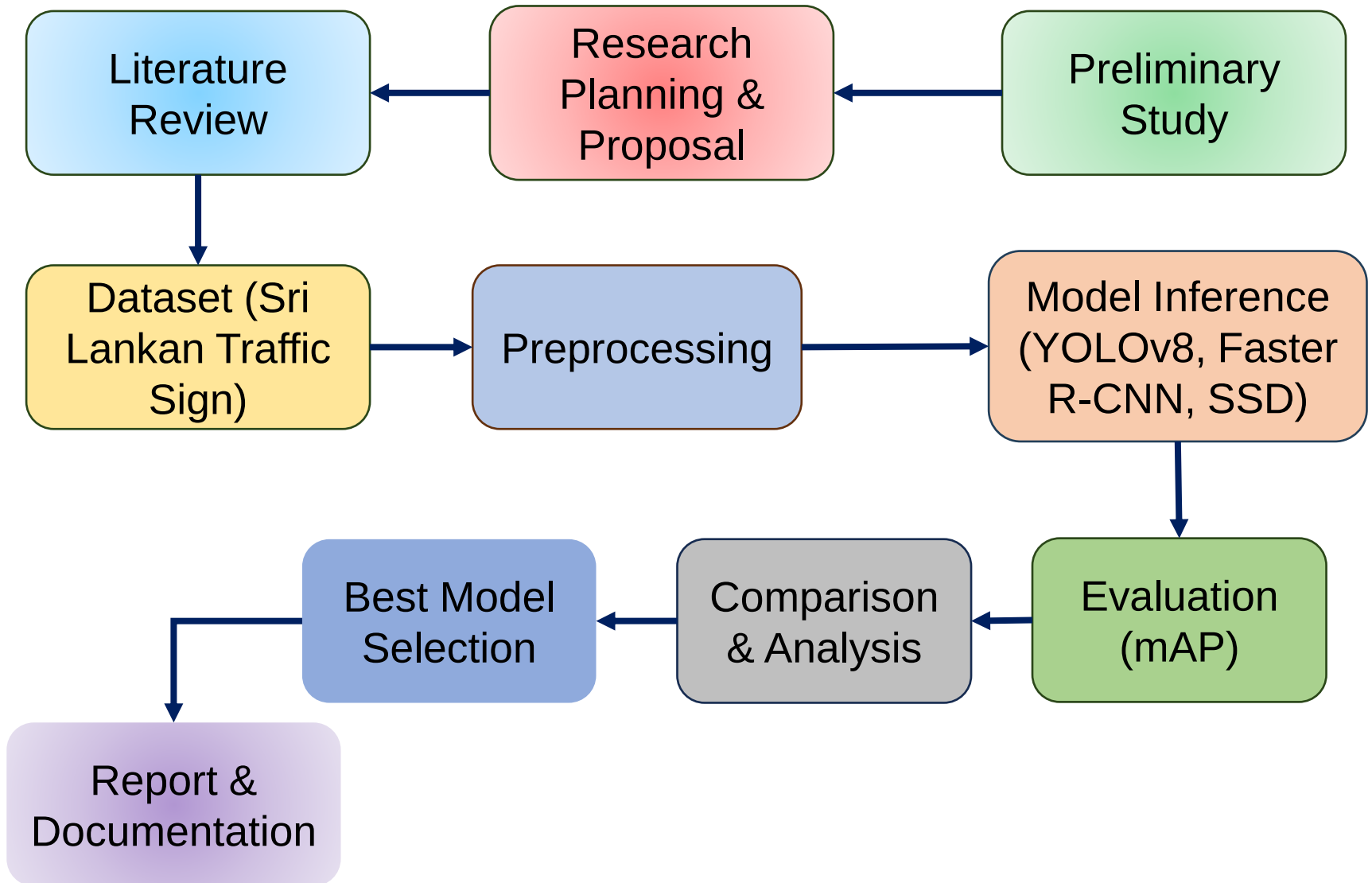
Benefits:

- Helps optimize model selection for real-world applications in Sri Lanka's transportation sector.
- Advances research in computer vision and AI for traffic management.

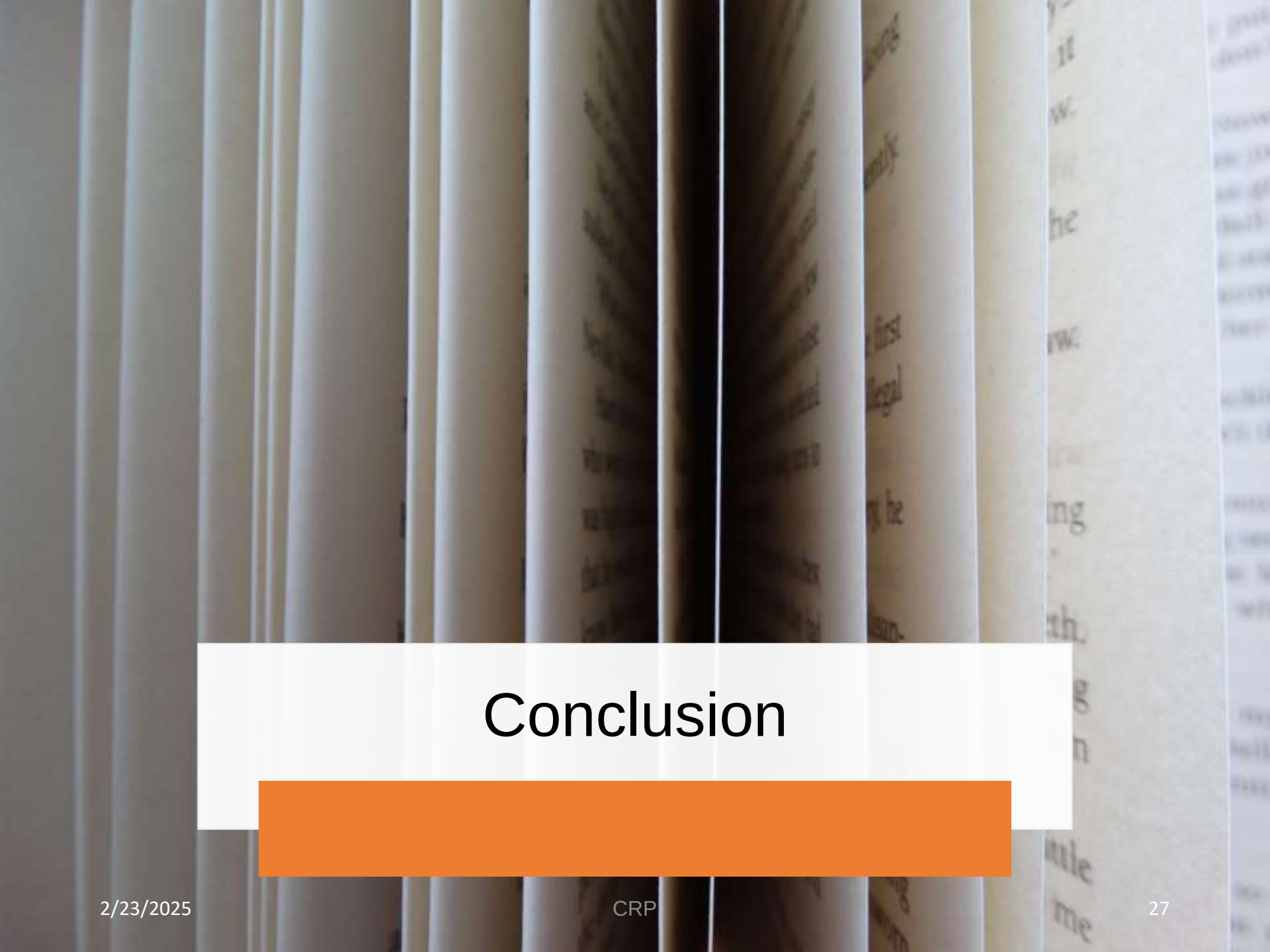


Top level architecture & WBS

Research Design



WBS	Task Description
1	Research Planning & Proposal
1.1	Define research problem & objectives
1.2	Prepare research proposal
1.3	Identify dataset and object detection models
2	Literature Review
2.1	Review related research on traffic sign detection
2.2	Study object detection techniques (YOLOv8, Faster R-CNN, SSD)
2.3	Analyze previous work on AI-based traffic systems
3	Data Collection & Preparation
3.1	Obtain the Sri Lankan traffic sign dataset
3.2	Perform data pre-processing
4	Model Evaluation & Analysis
4.1	Apply object detection models on the dataset
4.2	Compute mAP scores for each model
4.3	Compare and analyze model performance
5	Results Interpretation & Documentation
5.1	Interpret evaluation results
5.2	Document findings and insights
5.3	Discuss implications for AI-based traffic systems
6	Report Writing & Presentation
6.1	Write the final research report
6.2	Prepare the presentation slides
6.3	Present findings and conclusions

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Conclusion

Conclusion

- The study evaluates YOLOv8, Faster R-CNN, and SSD for Sri Lankan traffic sign detection.
- mAP (mean Average Precision) is used to measure model performance.
- The research identifies the most accurate model for Sri Lankan road conditions.
- Findings contribute to AI-driven traffic systems and autonomous vehicles.
- Helps improve road safety and smart transportation solutions in Sri Lanka.



Acknowledgement

Thanks To,

Supervisor: Mrs. NF. Mafaza.

Assessor: Mrs. ALF. Sajeetha.

Coordinator: Mr. MIM. Nismy.

BCAS Campus Kalmunai.



Questions?





Thank you